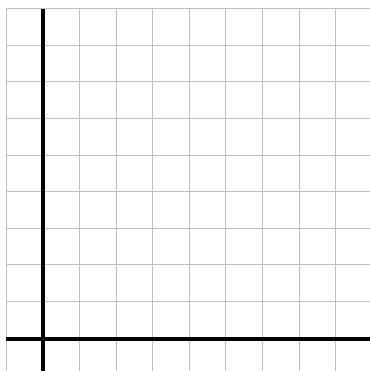
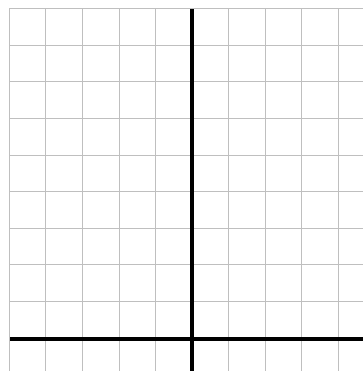


Parent Functions I: $\sqrt{x}$ and $|x|$ **TEACHER KEY** | Algebra II | Unit 1 | Day 4 | TEK A2.2.A**Objective:** I can graph the square root and absolute value parent functions from a table and analyze their key attributes: domain, range, intercepts, symmetry, and minimum.**Vocabulary****Parent function** — the **simplest** function in a family, with nothing added, subtracted, or multiplied.**x-intercept** — where the graph crosses the x-axis. There, $y = 0$. **y-intercept** — there, $x = 0$.**Symmetry** — the graph can be **folded or turned** onto itself.**Minimum** — the **smallest** y-value the function reaches.**Part 1 — The Square Root Function, $f(x) = \sqrt{x}$** *Fill in the table. Watch what happens with negative inputs.*

x	-4	-1	0	1	4	9
$f(x) = \sqrt{x}$	undef.	undef.	0	1	2	3

Why are the first two undefined? **you cannot take the square root of a negative number****Part 2 — The Absolute Value Function, $f(x) = |x|$** *Absolute value gives distance from zero, so the output is never negative.*

x	-5	-2	-1	0	1	2	5
$f(x) = x$	5	2	1	0	1	2	5

What shape does this make? **a V with its vertex at the origin****Part 3 — Graph Them Both***Each square is 1 unit. Plot your table values, then connect them smoothly.***A. $f(x) = \sqrt{x}$** (x from 0 to 9)**B. $f(x) = |x|$** (x from -5 to 5)Which quadrant only? **I**Fold line? **the y-axis****Part 4 — Compare the Key Attributes**

Attribute	$f(x) = \sqrt{x}$	$f(x) = x$
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Domain (interval)	$[0, \infty)$	$(-\infty, \infty)$
Range (interval)	$[0, \infty)$	$[0, \infty)$
x- and y-intercept	$(0, 0)$	$(0, 0)$
Symmetry	none	across the y-axis
Minimum	$y = 0$	$y = 0$
Increasing / decreasing	always increasing	down then up

Biggest difference: their **domains**. Only $\sqrt{x}$ refuses negative inputs.

Part 5 — Name That Function

Use only the attributes given. No equation.

- Domain and range both $[0, \infty)$, no symmetry, always increasing. $f(x) = \sqrt{x}$
- Domain all reals, range $[0, \infty)$, symmetric across the y-axis, minimum 0. $f(x) = |x|$

Practice

Evaluate, then answer. Show your work where there is room.

- Evaluate: $\sqrt{25} = 5$ $\sqrt{0} = 0$ $|-7| = 7$ $|0| = 0$
- Explain why $\sqrt{-9}$ is not a real number.

- Name every x-value where $|x| = 4$. $x = 4$ and $x = -4$ How many? **2**
- Name every x-value where $\sqrt{x} = 4$. $x = 16$ only How many? **1**
- Items 3 and 4 have different numbers of answers. What does that tell you about the two graphs?

- On $[1, 4]$, give the minimum and maximum of $f(x) = \sqrt{x}$. min **1** max **2**

Exit Ticket

Both functions have a minimum of 0 and pass through the origin. Give one attribute that tells them apart, and explain why it happens.

Attribute: **the domain**

Why: **$|x|$ accepts every real number, but $\sqrt{x}$ needs $x \geq 0$**